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ADJUSTABLE SHUNTS WITH IMPROVED FLOW CONTROL AND ASSOCIATED SYSTEMS AND METHODS

Abstract

The present technology is generally directed to adjustable shunting systems, including adjustable shunting systems with improved flow control. For example, the adjustable shunting systems described herein can include a flow control plate or other feature with a ramping feature that directs a sealing assembly toward and/or at least partially into an inlet or outlet of a drainage channel to improve the seal between the sealing assembly and the inlet or outlet. In some embodiments, the ramping feature is a slot, groove, or other opening in the plate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] The present application claims priority to U.S. Provisional Patent Application No. 63/338,393, filed May 4, 2022, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present technology generally relates to implantable medical devices and, in particular, to adjustable shunts for controlling fluid flow between a first body region and a second body region of a patient.

BACKGROUND

[0003] Implantable shunting systems are widely used to treat a variety of patient conditions by shunting fluid from a first body region/cavity to a second body region/cavity. For example, shunting systems have been proposed for treating glaucoma. The flow of fluid through the shunting systems is primarily controlled by the pressure gradient across the shunt and the physical characteristics of the flow path defined through the shunt (e.g., the resistance of the shunt lumen). Conventional, early shunting systems (sometimes referred to as minimally invasive glaucoma shunts or “MIGS”) have shown clinical benefit; however, there is a need for improved shunting systems and techniques for addressing elevated intraocular pressure and risks associated with glaucoma, as well as other patient conditions. For example, there is a need for shunting systems capable of adjusting the therapy provided, including the flow rate/fluid resistance between the two fluidly-connected bodies. As another example, there is a need for a shunting system able to be modified after manufacture (e.g., in the clinic) to personalize the system for the patient and/or as part of the clinician's plan for the implant procedure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Many aspects of the present technology can be better understood with reference to the following drawings. The components in the drawings are not necessarily drawn to scale. Instead, emphasis is placed on illustrating clearly the principles of the present technology. Furthermore, components can be shown as transparent in certain views for clarity of illustration only and not to indicate that the component is necessarily transparent. Components may also be shown schematically.

[0005] FIG. 1A illustrates a shunting system configured in accordance with select embodiments of the present technology.

[0006] FIG. 1B illustrates the shunting system of FIG. 1A rotated **180** degrees about its longitudinal axis.

[0007] FIG. 1C is an enlarged view of a portion of the shunting system shown in FIGS. 1A and 1B.

[0008] FIG. 2A illustrates an actuator for use with the shunting system shown in FIGS. 1A-1C and configured in accordance with select embodiments of the present technology.

[0009] FIGS. 2B-2D illustrate embodiments of the sealing assembly and gating element of the actuator shown in FIG. 2A and configured in accordance with select embodiments of the present technology.

[0010] FIG. 3 is a schematic illustration of a slot of the shunting system shown in FIGS. 1A-1C and a sealing assembly of the actuator shown in FIG. 2A.

[0011] FIGS. 4A1-4C1 illustrate a series of operating stages for adjusting flow through the shunting system of FIGS. 1A-1C in accordance with embodiments of the present technology, and FIGS. 4A2-4C2 illustrate another view of the series of operating stages for adjusting flow through the shunting system of FIGS. 1A-1C shown in FIGS. 4A1-4C1.

[0012] FIGS. 5 and 6 illustrate additional configurations of slots for use with shunting systems and configured in accordance with select embodiments of the present technology.

[0013] FIG. 7 is an exploded view of another shunting system configured in accordance with select embodiments of the present technology.

DETAILED DESCRIPTION

[0014] The present technology is generally directed to adjustable shunting systems, including adjustable shunting systems with improved flow control. For example, the adjustable shunting systems described herein can include a shunting element with at least one channel extending therethrough that permits fluid to flow through the system. The system can further include an actuator that can be selectively actuated to control the flow of fluid through an inlet or outlet of the channel to titrate the level of therapy provided by the shunt. In some embodiments, the actuator can include a gating element having a sealing assembly moveable between a first (e.g., closed) position in which the sealing assembly provides a first resistance to fluid flow through the inlet or outlet (e.g., by blocking the channel inlet or outlet), and a second (e.g., open) position in which the sealing assembly provides a second resistance less than the first resistance (e.g., by not blocking the channel inlet or outlet).

[0015] To further improve the flow control provided by the sealing assembly, the system can include a flow control plate that defines a ramping feature for directing the sealing assembly toward and/or at least partially into the inlet or outlet when the actuator is in the first position. As described throughout this Detailed Description, the ramping feature can be a slot, groove, or opening formed in the flow control plate and that extends at least partially between a first surface and a second opposite surface of the plate. The slot, for example, can be tapered such that it extends between a first relatively narrow end portion and a second relatively wider end portion (e.g., a width of the slot increases from the first relatively narrow end portion to the second relatively wider end portion). The tapered shape of the slot causes the sealing assembly to move in both a “horizontal” direction (e.g., parallel to a plane defined by the plate) and a “vertical” direction (e.g., perpendicular to the plane defined by the plate) as the sealing assembly moves between the first position and the second position. As described in detail throughout this Detailed Description, this multi-plane motion is expected to improve flow control through the system, such as by improving the seal obtained between the sealing assembly and the channel inlet or outlet when the sealing assembly is in the first position.

[0016] In addition to or in lieu of the flow control plate with the ramping feature, in some embodiments the system can include a moveable flap proximate the inlet or the outlet. In such embodiments, the actuator can direct the moveable flap into and/or over the inlet or outlet when the actuator (e.g., the sealing assembly) is in the first (e.g., closed) position. When the actuator (e.g., the sealing assembly) is in the second (e.g., open) position, the moveable flap generally does not cover the inlet or outlet. Accordingly, similar to the ramping feature, the moveable flap is expected to improve flow control through the system, such as by improving the seal obtained between the sealing assembly and the channel inlet or outlet when the sealing assembly is in the first position.

[0017] The terminology used in the description presented below is intended to be interpreted in its broadest reasonable manner, even though it is being used in conjunction with a detailed description of certain specific embodiments of the present technology. Certain terms may even be emphasized below; however, any terminology intended to be interpreted in any restricted manner will be overtly and specifically defined as such in this Detailed Description section. Additionally, the present technology can include other embodiments that are within the scope of the examples and claims but are not described in detail with respect to FIGS. 1A-6.

[0018] Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present technology. Thus, the appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features or characteristics may be combined in any suitable manner in one or more embodiments.

[0019] As used herein, the use of relative terminology, such as “about”, “approximately”, “substantially” and the like refer to the stated value plus or minus ten percent. For example, the use of the term “about **100**” refers to a range of from **90** to **110**, inclusive. In instances in which the context requires otherwise and/or relative terminology is used in reference to something that does not include a numerical value, the terms are given their ordinary meaning to one skilled in the art.

[0020] Reference throughout this specification to the term “resistance” refers to fluid resistance unless the context clearly dictates otherwise. The terms “drainage rate” and “flow rate” are used interchangeably to describe the movement of fluid through a structure at a particular volumetric rate. The term “flow” is used herein to refer to the motion of fluid, in general.

[0021] Although certain embodiments herein are described in terms of shunting fluid from an anterior chamber of an eye, one of skill in the art will appreciate that the present technology can be readily adapted to shunt fluid from and/or between other portions of the eye, or, more generally, from and/or between a first body region and a second body region. Moreover, while the certain embodiments herein are described in the context of glaucoma treatment, any of the embodiments herein, including those referred to as “glaucoma shunts” or “glaucoma devices” may nevertheless be used and/or modified to treat other diseases or conditions, including other diseases or conditions of the eye or other body regions. For example, the systems described herein can be used to treat diseases characterized by increased pressure and/or fluid build-up, including but not limited to heart failure (e.g., heart failure with preserved ejection fraction, heart failure with reduced ejection fraction, etc.), pulmonary failure, renal failure, hydrocephalus, and the like. Moreover, while generally described in terms of shunting aqueous, the systems described herein may be applied equally to shunting other fluid, such as blood or cerebrospinal fluid, between the first body region and the second body region.

[0022] FIGS. **1A-1C** illustrate an adjustable shunting system **100** (“the system **100**”) configured in accordance with select embodiments of the present technology. More specifically, FIG. **1A** is a perspective view of the system **100**, FIG. **1B** is a second perspective view of the system **100** with the system **100** rotated **180** degrees about its longitudinal axis relative to the view shown in FIG. **1A**, and FIG. **1C** is an enlarged view of a portion of the system **100** from the same view shown in FIG. **1B**. As described in greater detail below, the system **100** is configured to provide a titratable therapy for shunting fluid from a first body region to a second body region, such as shunting aqueous from an anterior chamber of a patient's eye to a target outflow location.

[0023] Referring collectively to FIGS. **1A** and **1B**, the system **100** includes a shunting element **102** and a plate or cartridge **110**. The shunting element **102** (which can also be referred to as a casing, membrane, elongated housing, or the like) extends between a first end portion **102a** and a second end portion **102b**. As best shown in FIG. **1A**, the shunting element **102** can have an opening **103** adjacent the first end portion **102a**. A plurality of flow channels **104** (shown as a first channel **104a**, a second channel **104b**, and a third channel **104c**) can extend through the shunting element **102** at least partially between the first end portion **102a** and the second end portion **102b**. The channels **104** can be fluidly isolated along a portion or substantial portion of the length of the shunting element **102**, and can merge into a primary drainage lumen **106** at or proximate the second end portion **102b** of the shunting element **102**. As described in greater detail below, when the system **100** is implanted within a patient between a first body region and a second body region, fluid can flow from the first body region to the second body region via the channels **104** and the primary drainage lumen **106**. The shunting element **102** may optionally include one or more features to

facilitate anchoring the system **100** to patient tissue, such as first and second suture holes **108a**, **108b**. Additional features of shunting elements suitable for use with the present technology are described in International Patent Application No. PCT/US2022/037747, the disclosure of which is incorporated by reference herein in its entirety and for all purposes.

[0024] The plate **110** (which can also be referred to as a flow control plate, a flow control cartridge, a flow control feature, etc.) can be positioned within the first end portion **102a** of the shunting element **102**. As best shown in FIG. 1A (which shows a first or upper surface **110a** of the plate **110**), the plate **110** can partially align with the opening **103** of the shunting element **102**. More specifically, the plate **110** can be positioned within the shunting element **102** such that a first port or opening **112a**, a second port or opening **112b**, and a third port or opening **115** in the plate **110** align with the opening **103**. In operation, fluid can enter the system **100** by flowing through the opening **103** and the first port **112a**, the second port **112b**, and/or the third port **112c**.

[0025] The plate **110** can further include ramping features comprising a first slot **114a** and a second slot **114b** (collectively referred to herein as “the slots **114**”). Each of the slots **114** can be a groove, opening, or other slit that forms an indentation or through-hole in the plate **110**. In other embodiments, the slots **114** can be formed by one or more ridges of equal or substantially equal height that extend from a surface of the plate **110**. In such embodiments, the slots **114** comprise the “empty space” or chamber formed between the one or more ridges. Regardless of how they are formed, the slots **114** can have a tear-drop shape or other similar shape, such that a “width” of each slot **114** is tapered along the length of each slot **114**. As described in greater detail with respect to FIGS. 4A1-4C2, the slots **114** act as a ramp to assist with improving a seal formed between inlets to the channels **104** and a corresponding actuator configured to selectively control flow through the inlets (described with respect to FIG. 2; not shown in FIGS. 1A-1C). In the illustrated embodiment, the slots **114** also align with the opening **103** in the shunting element **102** such that, in embodiments in which the slots **114** extends through the full thickness of the plate **110**, fluid may also flow into the system **100** via the slots **114**. In other embodiments, however, the slots **114** are not aligned with the opening **103** and do not facilitate fluid flow into the system **100**, and instead primarily provide the ramping feature. Regardless, an outer surface of the plate **110** (e.g., the upper surface **110a**) generally forms a substantial fluid seal with an inner surface of the shunting element **102** such that fluid does not leak therebetween. As a result, fluid only enters the system **100** via the first port **112a**, the second port **112b**, and the third port **115** (and the slots **114** in embodiments in which they align with the opening **103** and extend through the full thickness of the plate **110**).

[0026] As shown in FIG. 1B (which shows a second or lower surface **110b** of the plate **110**), the plate **110** can further include a first actuator chamber **116a** and a second actuator chamber **116b** (collectively referred to as “the actuator chambers **116**”). The first actuator chamber **116a** is fluidly coupled to the first port **112a** and the first slot **114a** such that fluid flowing through the first port **112a** and/or the first slot **114a** enters the first actuator chamber **116a**. Likewise, the second actuator chamber **116b** is fluidly coupled to the second port **112b** and the second slot **114b** such that fluid flowing through the second port **112b** and/or the second slot **114b** enters the second actuator chamber **116b**. In some embodiments, the actuator chambers **116** are fluidly isolated. In other embodiments, the actuator chambers **116** are fluidly connected. Regardless, the actuator chambers **116** can be configured to receive corresponding actuators (not shown in FIG. 1B; a representative actuator is illustrated in, and described in detail with respect to, FIG. 2) for selectively controlling the flow of fluid through the system **100**, as described in greater detail below.

[0027] As also shown in FIG. 1B, the channels **104** extend proximate the first end portion **102a** of the shunting element **102** such that the plate **110** and the channels **104** at least partially overlap and are in fluid communication. In particular, the first channel **104a** overlaps with (and/or is otherwise aligned with) the first actuator chamber **116a**, the second channel **104b** overlaps with the second actuator chamber **116b**, and the third channel **104c** interfaces with the third port **115**. As described in greater detail with respect to FIG. 1C, the first channel **104a** is fluidly coupled with and can be

configured to receive fluid from the first actuator chamber **116a** via a first aperture or inlet (not shown) and the second channel **104b** is fluidly coupled with and can be configured to receive fluid from the second actuator chamber **116b** via a second aperture or inlet (not shown). The actuators (not shown) can be configured to selectively and independently interface with the corresponding first and second apertures to control the flow of fluid through the first channel **104a** and the second channel **104b**.

[0028] FIG. **1C** is an enlarged view of the first actuator chamber **116a** and illustrates additional details of the system **100**. In particular, FIG. **1C** illustrates the first aperture or inlet **105a** that fluidly connects the first actuator chamber **116a** and the first channel **104a**. That is, fluid can flow from the first actuator chamber **116a** into the first channel **104a** via the first aperture **105a**. As illustrated, the first aperture **105a** is aligned with (e.g., vertically aligned with) at least a portion of the first slot **114a**. In particular, the first aperture **105a** is aligned with a first portion **114a1** of the first slot that is relatively narrow (e.g., as opposed to being aligned with a second portion **114a2** of the first slot **114a** that is relatively wider than the first portion **114a1**). As described in detail with respect to FIGS. **4A1-4C2**, aligning the first aperture **105a** with the relatively narrow portion **114a1** of the first slot **114a** improves the sealing of the first aperture **105a** when the actuator (not shown) is transitioned to a “closed” position in which it blocks or substantially blocks fluid from flowing into the first channel **104a** via the first aperture **105a**. In some embodiments, such as those described in greater detail with reference to FIG. **7**, the system **100** can include a hinged or otherwise moveable feature coupled proximate to the first aperture **105a** that can also improve the sealing of the first aperture **105a** when the actuator is transitioned to the closed position.

[0029] As also illustrated in FIG. **1C**, the first actuator chamber **116a** can also include a first anchoring chamber **120a**, a second anchoring chamber **121a**, and a third anchoring chamber **122a**. As described in greater detail below with reference to FIG. **2**, the anchoring chambers **120a-122a** are configured to receive and retain corresponding anchoring elements of an actuator such that the actuator is retained within the first actuator chamber **116a**.

[0030] FIG. **2A** illustrates a representative actuator **200** that can be used with the system **100** and is configured in accordance with select embodiments of the present technology. The actuator **200** includes a projection or gating element **232** having a distal end portion **232a** configured to at least partially control (e.g., gate) flow through the system **100**. To do so, the distal end portion **232a** includes a sealing assembly **234** having a first portion **234a** positioned on a first (e.g., upper) surface of the distal end portion **232a** and a second portion **234b** positioned on a second (e.g., lower) surface of the distal end portion **232a**. As described in detail below, the first portion **234a** is configured to slidably engage the first slot **114a**, and can therefore be referred to as a ramping element or feature **234a**, while the second portion **234b** is configured to releasably engage the first aperture **105a**, and can therefore be referred to as a sealing element or feature **234b**. As also described in detail below, the sealing assembly **234** is expected to improve the flow-blocking effect of the gating element **232**.

[0031] The sealing assembly **234** and the distal end portion **232a** of the gating element **232** can be designed to interface in a variety of different manners. FIGS. **2B-2D** are cross-sectional views of different embodiments of the sealing assembly **234** and distal end portion **232a** of the gating element **232** taken along the line **2B-2D** shown in FIG. **2A**. For ease of reference, FIGS. **2B-2D** use the same primary reference numbers to refer to corresponding components shown in FIG. **2A**, but add an additional subscript numeral to denote that each component may not be identical in the various embodiments (e.g., in FIG. **2B**, the sealing assembly **234** is referred to as the sealing assembly **234.sub.1**; in FIG. **2C**, the sealing assembly **234** is referred to as the sealing assembly **234.sub.2**; etc.).

[0032] As shown in FIG. **2B**, in some embodiments the first portion **234a.sub.1** and the second portion **234b.sub.1** of the sealing assembly **234.sub.1** are discrete components. In such embodiments, the first portion **234a.sub.1** can be coupled to an upper surface of the distal end

portion **232a.sub.1** of the gating element **232** via mechanical tethers, bands, barbs, staples, sutures, glue, chemical bonding, or the like, and the second portion **234b.sub.1** can be separately coupled to a lower surface of the distal end portion **232a.sub.1** via mechanical tethers, bands, barbs, staples, sutures, glue, chemical bonding, or the like. Although shown as having a semi-spherical shape, in other embodiments the first portion **234a.sub.1** and/or the second portion **234a.sub.2** can have other suitable shapes, such as a conical shape.

[0033] In other embodiments, the first portion **234a.sub.1** and the second portion **234b.sub.1** are a single unitary component. For example, the first portion **234a.sub.1** and the second portion **234b.sub.1** of the sealing assembly **234.sub.1** can be connected by a medial portion (not shown) that extends through an aperture in the distal end portion **232a.sub.1** of the gating element **232** and that secures the sealing assembly **234.sub.1** to the distal end portion **232a.sub.1**. As another example, the first portion **234a.sub.1** and the second portion **234b.sub.1** can be a coating or other material on the surface of the gating element **232** (e.g., completely or at least partially surrounding the distal end portion **232a.sub.1**). As yet another example, the sealing assembly **234.sub.1** itself can form the distal end portion **232a.sub.1** of the gating element **232** such that the sealing assembly **234.sub.1** is integral with the gating element **232**. In such embodiments, the sealing assembly **234.sub.1** can be generally spherical or conical in shape, with the first portion **234a.sub.1** being a first hemispherical portion of the sphere and the second portion **234b.sub.1** being a second hemispherical portion of the sphere.

[0034] In some embodiments, the sealing assembly **234** or a portion thereof is movable relative to (e.g., configured to rotate relative to) the distal end portion **232a** of the gating element **232**. For example, as shown in FIG. 2C, the sealing assembly **234.sub.2** can have a spherical shape and be positioned within a corresponding cavity or opening **233.sub.2** in the distal end portion **232a.sub.2**. A first hemispherical portion **234a2** of the sealing assembly **234.sub.2** forms the “first portion **234a**” of the sealing assembly **234.sub.2**, and a second hemispherical portion **234b.sub.2** of the sealing assembly **234.sub.2** forms the “second portion **234b**” of the sealing assembly **234.sub.2**. The sealing assembly **234.sub.2** can be rotatably contained within the distal end portion **232a.sub.2**, such that as the distal end portion **232a.sub.2** is translated in a first direction indicated by arrow A, the sealing assembly **234.sub.2** rotates in a second direction indicated by arrow B. Enabling rotation of the sealing assembly **234.sub.2** relative to the distal end portion **232a.sub.2** may improve performance of the actuator **200** because it permits the sealing assembly **234.sub.2** to rotate as it slidably engages the first slot **114a**, reducing the frictional interference between the sealing assembly **234.sub.2** and the first slot **114a**.

[0035] FIG. 2D illustrates another representative embodiment of a sealing assembly **234.sub.3** that is movable relative to (e.g., configured to rotate relative to) a distal end portion **232a.sub.3** of the gating element **232**. In the illustrated embodiment, the distal end portion **232a.sub.3** of the gating element **232** forms a cup or recess **233.sub.3** that houses the sealing assembly **234.sub.3**. The sealing assembly **234.sub.3** is rotatably moveable relative to the recess **233.sub.3**, such that as the distal end portion **232a.sub.3** is translated in a first direction indicated by arrow A, the sealing assembly **234.sub.3** rotates within the recess **233.sub.3** in a second direction indicated by arrow B. In the illustrated embodiment, the first portion **234a.sub.3** is configured to slidably engage the first slot **114a**. However, the second portion **234b.sub.3** sits within the recess **233.sub.3** and therefore does not directly engage with the first aperture **105a**. Rather, a lower surface **232a.sub.3x** of the distal end portion **232a.sub.3** is configured to releasably engage the first aperture **105a**.

[0036] Returning to FIG. 2A, and regardless of the specific embodiment of the sealing assembly **234** and the distal end portion **232a**, the sealing assembly **234** can be at least partially composed of an impermeable and partially compressible or flexible material (e.g., an elastomeric material, a material having a low durometer, a material having a low Young's modulus, etc.). For example, in some embodiments at least the second portion **234b** (i.e., the sealing element **234b**) is composed of the impermeable and partially compressible material. As described in greater detail below, this

enables the second portion **234b** to conform to a corresponding channel inlet/aperture to control the flow therethrough. The sealing assembly **234** can also or alternatively be at least partially composed of a generally rigid material (e.g., a non-compressible or non-flexible material having a generally high durometer and/or high Young's modulus). For example, in some embodiments at least the first portion **234a** (i.e., the ramping element **234a**) is composed of a rigid material that does not substantially deform. Examples of suitable materials having a generally high durometer include, but are not limited to, glass, titanium, Nitinol, stainless steel, high-durometer silicone, high-durometer polymers, or the like. As described in greater detail below, this enables the first portion **234a** to slide along the slot **114a** without substantially deforming. In some embodiments, both the first portion **234a** and the second portion **234b** are composed of the rigid material that does not substantially deform.

[0037] The sealing assembly **234** can have a dimension (e.g., a height, width, diameter, etc.) that is between about 25 microns and 200 microns, or between about 50 microns and 175 microns, or between about 70 microns and about 150 microns. In some embodiments, a largest dimension (e.g., whichever “edge-to-edge” dimension of height, width, or diameter is greatest) is less than about 150 microns, less than about 125 microns, less than about 100 microns, less than about 90 microns, less than about 80 microns, or less than about 70 microns. The foregoing ranges are provided by way of example only—in some embodiments, the sealing assembly **234** can have dimensions outside of the foregoing ranges.

[0038] The actuator **200** further includes a first actuation element **238a** and a second actuation element **238b**. The first actuation element **238a** can be configured to rotate, pivot, slide, or otherwise move the gating element **232**, and thus the sealing assembly **234**, in a first direction. The second actuation element **238b** can be configured to selectively rotate, pivot, slide, or otherwise move the gating element **232**, and thus the sealing assembly **234**, in a second direction generally opposite the first direction. For example, the first actuation element **238a** and the second actuation element **238b** can be composed at least partially of a shape memory material or alloy (e.g., nitinol). Accordingly, the first actuation element **238a** and the second actuation element **238b** can be transitionable at least between a first material phase or state (e.g., a martensitic state, a R-phase, a composite state between martensitic and R-phase, etc.) and a second material phase or state (e.g., an austenitic state, an R-phase state, a composite state between austenitic and R-phase, etc.). In the first material state, the first actuation element **238a** and the second actuation element **238b** may have reduced (e.g., relatively less stiff) mechanical properties that cause the actuation elements to be more easily deformable (e.g., compressible, expandable, etc.) relative to when the actuation elements are in the first material state. In the second material state, the first actuation element **238a** and the second actuation element **238b** may have increased (e.g., relatively more stiff) mechanical properties relative to the first material state, causing an increased preference toward a specific preferred geometry (e.g., original geometry, manufactured or fabricated geometry, heat set geometry, etc.). The first actuation element **238a** and the second actuation element **238b** can be selectively and independently transitioned between the first material state and the second material state by applying energy (e.g., laser energy, electrical energy, etc.) to the first actuation element **238a** or the second actuation element **238b** to heat it above a transition temperature (e.g., above an austenite finish (Af) temperature, which is generally greater than body temperature). If the first actuation element **238a** (or the second actuation element **238b**) is deformed relative to its preferred geometry when heated above the transition temperature, the first actuation element **238a** (or the second actuation element **238b**) will move to and/or toward its preferred geometry. In some embodiments, the first actuation element **238a** and the second actuation element **238b** are operably coupled such that, when the actuated actuation element (e.g., the first actuation element **238a**) transitions toward its preferred geometry, the non-actuated actuation element (e.g., the second actuation element **238b**) is further deformed relative to its preferred geometry. Additional details regarding, and examples of, bi-directional shape memory actuators are described in U.S. Patent

Application Publication Nos. 2020/0229982 and 2021/0251806, the disclosures of which are incorporated by reference herein in their entireties and for all purposes.

[0039] The actuator **200** further includes a first anchoring element **240**, a second anchoring element **241**, and a third anchoring element **242**. To couple the actuator **200** to the system **100**, the first anchoring element **240** of the actuator **200** can be secured within the first anchoring chamber **120a** (FIG. **1C**) of the first actuator chamber **116a**, the second anchoring element **241** can be secured within the second anchoring chamber **121a** (FIG. **1C**) of the first actuator chamber **116a**, and the third anchoring element **242** can be secured within the third anchoring chamber **122a** (FIG. **1C**) of the first actuator chamber **116a**. In some embodiments, the first actuation element **238a** and the second actuation element **238b** are deformed relative to their preferred geometries (e.g., “loaded”) when the actuator **200** is positioned within the corresponding actuator chamber of the system **100** (e.g., the first actuator chamber **116a**; FIG. **1C**). For example, the anchoring chambers **120a-122a** can be configured/dimensioned such that the act of placing the corresponding anchoring elements **240-242** within the anchoring chambers **120a-122a** deforms the first actuation element **238a** and the second actuation element **238b** relative to their original manufactured geometries. In some embodiments positioning the anchoring elements **240-242** within corresponding anchoring chambers **120a-122a** can increase a length of the actuation elements **138** (e.g., tension) relative to their preferred geometries. In other embodiments, positioning the anchoring elements **240-242** within corresponding anchoring chambers **120a-122a** can decrease a length of the actuation elements **138** (e.g., compress) relative to their preferred geometries. Additional details regarding loading and deforming shape memory actuators are described in U.S. Patent Application Publication No. US 2021/0251806, previously incorporated by reference herein, and International Patent Application No. PCT/US21/49140, the disclosure of which is incorporated by reference in its entirety and for all purposes.

[0040] The distal end portion **232a** of the gating element **232** is configured to moveably interface with various features of the system **100** to at least partially control the flow of fluid through one or more flow pathways extending through the system **100**. For example, referring collectively to FIGS. **1C** and **2A**, when the actuator **200** is positioned within the first actuator chamber **116a**, the distal end portion **232a** of the gating element **232** is positioned generally between the first aperture **105a** and the first slot **114a**. As a result, the first portion **234a** (i.e., the ramping element **234a**) of the sealing assembly **234** is configured to slidably engage the first slot **114a**, and the second portion **234b** (i.e., the sealing element **234b**) of the sealing assembly **234** is configured to releasably engage the first aperture **105a**.

[0041] The actuation elements **238** are configured to translate the sealing assembly **234** in a generally “horizontal” direction along a length of the slots **114** (e.g., within a plane parallel to a plane defined by the plate, labeled in FIG. **1C** as the axis X). Depending on the configuration of the actuator **200**, the actuation elements **238** may translate the sealing assembly **234** along an arc as opposed to along a linear path (e.g., in embodiments in which the actuation elements **238** cause the gating element **232** to rotate or pivot). Such translation still occurs within the plane parallel to the plane defined by plate, just along an arcuate path instead of a linear path. In such embodiments, the slot **114** can be at least partially curved (e.g., a curved tear-drop shape) such that the slot **114** traces the sealing assembly's translation arc.

[0042] Additionally, as illustrated and described below with reference to FIGS. **4A1-4C2**, as the sealing assembly **234** is horizontally translated via actuation of the actuation elements **238**, the slots **114** act as “two-dimensional ramps” that change a “vertical” position of the sealing assembly **234** (e.g., within a plane normal to the plane defined by the plate, labeled in FIG. **1C** as the axis Y). In particular, the variable/tapered width of the slots **114** affect how much of the sealing assembly **234** can “fit” within the slot **114**, and thus affect a vertical position of the sealing assembly **234**. For example, FIG. **3** schematically illustrates a top view of the first slot **114a** and a side view of the sealing assembly **234**. As seen, the second portion **114a.sub.2** of the slot **114** has a first width A that

is equal or approximately equal to a first dimension A (i.e., the diameter) of the sealing assembly **234**, a middle portion of the first slot **114a** has a second width B that is less than the first width A and is equal or approximately equal to a second dimension B of the sealing assembly **234**, and the first portion **114a.sub.1** of the first slot **114a** has a third width C that is less than the first width A and the second width B, and that is equal or approximately equal to a third dimension C of the sealing assembly **234**. As a result, when the sealing assembly **234** is aligned with the second, relatively wider, portion **114a.sub.2**, nearly half of the sealing assembly **234** can fit within the slot **114a**. However, when the sealing assembly **234** is aligned with the first, relatively narrower, portion **114a.sub.1** of the first slot **114a**, only a small portion of the sealing assembly **234** can fit within the first slot **114a**. To prevent the sealing assembly **234** from fully “falling into” or otherwise fitting within the first slot **114a**, the largest dimension of the sealing assembly **234** (e.g., the largest diameter) can be at least slightly larger than the largest dimension (e.g., width) of the first slot **114a**.

[0043] Accordingly, referring collectively to FIGS. **1C-3**, as the sealing assembly **234** moves from the second, relatively wider, portion **114a.sub.2** of the first slot **114a** toward the first, relatively narrow, portion **114a.sub.1** of the first slot **114a** (e.g., via actuation of the first actuation element **238a**), less of the first portion **234a** of the sealing assembly **234** can fit within the first slot **114a**. This causes the sealing assembly **234** to be pushed “out of” the first slot **114a** in a first vertical direction perpendicular to the plane of the plate **110** in which the first slot **114a** is formed (e.g., in a direction vertically away from the slot **114a**, such as along the axis Y labeled in FIG. **1C**). And because the first, relatively narrow, portion **114a.sub.1** of the first slot **114a** is vertically aligned with the first aperture **105a**, this first vertical movement of the sealing assembly **234** away from slot **114a** causes the second portion **234b** of the gating element **232** to move toward and/or at least partially into the first aperture **105a**, thereby improving the fluidic seal formed between the gating element **232** and the first aperture **105a**. The slot **114a** can therefore direct the second portion **234b** of the gating element **232** to abut, conform to, and/or be partially inserted into the first aperture **105a**.

[0044] In reverse, as the sealing assembly **234** moves from the first, relatively narrower, portion **114a.sub.1** of the slot **114a** toward the second, relatively wider, portion **114a.sub.2** of the first slot **114a** (e.g., via actuation of the second actuation element **238b**), more of the first portion **234a** of the sealing assembly **234** can fit within the first slot **114a**. This causes the sealing assembly **234** to move in a second direction perpendicular to the plane of the plate (e.g., in a direction vertically toward/into the first slot **114a**, such as also along the axis Y labeled in FIG. **1C**) that is opposite the first direction described above. This increases the vertical distance between the second portion **234b** of the sealing assembly **234** and the aperture **105a**, causing the sealing assembly **234** to disengage from the first aperture **105a** and permit fluid to flow therethrough.

[0045] Accordingly, the sealing assembly **234** moves in two directions during actuation: (1) horizontally/laterally relative to the plate **110** (e.g., along the axis X shown in FIG. **1C**), caused by actuation of a corresponding actuation element **138**, and (2) vertically/perpendicular to the plane of the plate **110** (e.g., along the axis Y shown in FIG. **1C**), caused by the sealing assembly **234** falling into or climbing out of the first slot **114a**. Accordingly, the net motion of the sealing assembly **234** is angled relative to the plate **110** by virtue of the first slot **114a**. In this way, the first slot **114a** acts as a “two-dimensional ramp,” despite the first slot **114a** not actually having a surface that is sloped relative to the plate **110**.

[0046] FIGS. **4A1-4C1** are a series of top views illustrating movement of the first portion **234a** (i.e., the ramping element **234a**) of the sealing assembly **234** relative to the first slot **114a**, and FIGS. **4A2-4C2** are a series of slightly offset and elevated end views corresponding to the top views shown in FIGS. **4A1-4C1**, respectively. In particular, FIGS. **4A1** and **4A2** illustrate a first position in which the first portion **234a** of the sealing assembly **234** is aligned with the second, relatively wider, portion **114a.sub.2** of the first slot **114a**. FIGS. **4B1** and **4B2** illustrate a second

(e.g., intermediate) position as the sealing assembly **234** moves from the second, relatively wider portion **114a.sub.2** toward the first, relatively narrower, portion **114a.sub.1** of the first slot **114a**, such as during actuation of the first actuation element **238a**. FIGS. **4C1** and **4C2** illustrate a third position in which the first portion **234a** of the sealing assembly **234** is aligned with the first, relatively narrower, portion **114a.sub.1** of the first slot **114a**.

[0047] In the first position (FIGS. **4A1** and **4A2**), the first portion **234a** of the sealing assembly **234** can sit at least partially within the first slot **114a**, which causes the sealing assembly **234** to be positioned vertically away from the first aperture **105a** (best shown in FIG. **4A2**). However, as the sealing assembly **234** is moved toward the third position shown in FIGS. **4C1** and **4C2** (i.e., from the second portion **114a.sub.2** of the slot **114** toward the first portion **114a.sub.1** of the slot **114**), less of the first portion **234a** of the sealing assembly **234** “fits” within the first slot **114a**. This causes the sealing assembly **234** to be pushed vertically away from the first slot **114a** as it moves from the first position toward the third position. As previously described, the first aperture **105a** is vertically aligned with the first portion **114a.sub.1** of the first slot **114a**. Thus, the vertical movement of the sealing assembly **234** caused by the first slot **114a** as the sealing assembly **234** moves from the first position toward the third position directs the sealing assembly **234** (and more specifically, the second portion **234b** of the sealing assembly) into the first aperture **105a**, thereby improving the seal between the sealing assembly **234** and the first aperture **105a** (for purposes of illustration, the second portion **234b** is shown as slightly spaced apart from the first aperture **105a**; one skilled in the art will appreciate from the disclosure herein that the second portion **234b** can abut and even at least partially sit within the first aperture **105a** when in the third position illustrated in FIGS. **4C1** and **4C2**). The operation can be reversed to “open” the first aperture **105a** (e.g., the sealing assembly **234** can be translated from the third position toward the first position) by actuating the other actuation element.

[0048] Flow through the second channel **104b** can be controlled in the same or generally similar manner as flow through the first channel **104a**. For example, the system **100** can include a second actuator **200** positioned in the second actuator chamber **116b** that selectively interferes with a second aperture or inlet to the second channel **104b**. The second slot **114b** can perform the same or similar function as described above with respect to the first slot **114a**, but relative to the second aperture or inlet. That is, the second slot **114b** can act as a two-dimensional ramp to improve flow control through the second channel **104b**. In contrast to the first channel **104a** and the second channel **104b**, the third channel **104c** is designed to be “always open” such that it permits at least some degree of fluid flow through the system **100** even when both the first channel **104a** and the second channel **104b** are blocked/closed. Of course, the present technology is not limited to particular combinations of “always open” and adjustable channels, and can include more or fewer of each channel type.

[0049] Moreover, although FIGS. **1A-4C2** show and describe the slot **114a** as being “above” the actuator **200** and the channel inlet (i.e., the first aperture **105a**) as being “below” the actuator **200**, the relative positioning of these features can be reversed, such that the slot **114a** is “below” the actuator **200** and the channel inlet (i.e., the first aperture **105a**) is “above” the actuator. In such embodiments, the slot **114a** can be downstream of the first aperture **105a** (e.g., as opposed to being upstream of the first aperture **105a**, as described with reference to FIGS. **1A-4C2**).

[0050] The slots described herein can have other suitable shapes and configurations while still acting as a “two-dimensional” ramp. For example, FIG. **5** illustrates a slot **514** configured in accordance with select embodiments of the present technology. Similar to the slot **114**, the slot **514** extends between a first, relatively narrow, end portion **514a** and a second, relatively wider, end portion **514b**. However, unlike the slot **114** which is formed by two side walls that are relatively straight, the slot **514** is formed in part by two partially curved side walls **515**. As a result, the width of the slot **514** does not increase linearly between the first end portion **514a** and the second end portion **514b**. As one skilled in the art will appreciate from the disclosure herein, changing the

tapering of the slot **514** will change the vertical vs. horizontal movement of the sealing assembly (not shown) as it moves between the first end portion **514a** and the second end portion **514b**. For example, the shape of the slot **514** can be manipulated to control the relative degree of horizontal versus vertical translation. In some embodiments, the sealing assembly (not shown) can have an irregular shape or curvature that is designed to control the relative degree of horizontal versus vertical translation, in addition to or in lieu of the slot **514** being manipulated to control the relative degree of horizontal versus vertical translation.

[0051] FIG. **6** illustrates another slot **614** configured in accordance with select embodiments of the present technology. The slot **614** is generally similar to the slots **114**, but further includes one or more frictional or resistive elements **615**. The frictional or resistive elements **615** can be ridges, bumps, projections, or the like. Alternatively, the frictional elements **615** can be a recess or other widening in the width of the slot **614**. In operation, the frictional elements **615** at least partially restrict movement of the sealing assembly (not shown) as it moves between the first end portion **614a** and the second end portion **614b**. However, the frictional elements **615** do not interfere with the sealing assembly to the extent that it is permanently prevented from moving between the first end portion **614a** and the second portion **614b**. Rather, the frictional elements **615** reduce or prevent unwanted movement of the sealing assembly, but permit the sealing assembly to move in response to the corresponding actuator (not shown) coupled to the sealing assembly being actuated. In other words, moving the sealing assembly **234** past the friction elements **615** requires an energy input (e.g., via actuation of the actuator) and helps prevent and/or at least reduce unwanted or passive movement of the sealing assembly **234**. Examples of using frictional elements to improve the operation of adjustable shunting systems are described in International Patent Application No. PCT/US21/49140, the disclosure of which was previously incorporated by reference herein.

[0052] Without being bound by theory, the slots described herein (e.g., the slots **114**, **514**, and **614**) are expected to improve performance of adjustable shunting systems, such as the system **100**. In particular, the slots are expected to reduce or minimize leakage of fluid through the corresponding channel inlet (e.g., the first aperture **105a**) when a user sets the actuator to the “closed” position. That is, the slots improve the degree of control over fluid flow through adjustable shunting systems by providing a better “seal” when the actuator is moved to the “closed” position. In some embodiments, however, a small, fixed leak through the corresponding inlet may still exist even when the actuator has been moved to the “closed” position.

[0053] An additional expected advantage of the present technology is that, in certain embodiments and depending on the manufacturing methods utilized to make the system, the slots are expected to be relatively easier to manufacture than “three-dimensional” ramps that have a sloped surface extending from the plate. For example, removing material from the plate to form the slot can be simpler than adding material to form a sloped three-dimensional ramp. Of course, the slots and other ramping features described herein provide other advantages beyond those expressly described herein, and the present technology is therefore not limited by the foregoing expected advantages. Similarly, the slots or other similar ramping features may be incorporated into adjustable shunting systems other than those expressly described herein.

[0054] The systems described herein can have other features expected to improve the sealing ability of the actuator, in combination with or in lieu of the slots **114**. For example, FIG. **7** is an exploded view of another adjustable shunting system (“the system **700**”) configured in accordance with select embodiments of the present technology. The system **700** can include certain features that are generally similar to some of the features of the system **100** described above. Accordingly, the description below generally focuses on features of the system **700** that differ from the system **100**. However, one skilled in the art will appreciate from the disclosure herein that various aspects of the system **100** can be incorporated into the system **700**, and vice versa.

[0055] The system **700** includes a shunting element **702** comprising a first layer **702a**, a second layer **702b**, and a third layer **702c**. When assembled, a lower surface of the first layer **702a** is

coupled (e.g., adhered, bonded, etc.) to an upper surface of the second layer **702b**, and a lower surface of the second layer **702b** is coupled to the upper surface of the third layer **702c**. The third layer **702c** defines a plurality of channels **704** for permitting fluid to flow through the system **700**, similar to the channels **104** of the system **100**. Additional details regarding multi-layered shunting elements are described in International Patent Application Publication No. WO 2023/004067, the disclosure of which is incorporated by reference herein in its entirety.

[0056] The system **700** further includes a first plate or cartridge **710**. The first plate **710** can be generally similar to the plate **110** described with reference to FIGS. 1A-1C. For example, the first plate **710** can include a first chamber **716a** for receiving a first actuator **730a** and a second chamber **716b** for receiving a second actuator **730b**. Relative to the plate **110** of FIGS. 1A-1C, the first plate **710** does not necessarily include the slot **114**. Instead, as described below, the system **700** has a different mechanism for improving the sealing-capability of the first actuator **730a** and the second actuator **730b**. The system **700** can optionally include a second plate **715** that forms a “top” to the first plate **710** and holds the first actuator **730a** and the second actuator **730b** in the corresponding chambers **716a/b** of the first plate **710**. The second plate **715** includes a plurality of apertures **712** that permit fluid to flow through the second plate **715** (e.g., toward the channels **704**).

[0057] The first actuator **730a** and the second actuator **730b** (collectively referred to herein as “the actuators **730**”) can be generally similar to the actuator **200** described with reference to FIGS. 2A-2D. For example, the first actuator **730a** can include a first gating element **732a** and the second actuator **730b** can include a second gating element **732b** (collectively, “the gating elements **732**”). However, in some embodiments the actuators **730** do not have a “sealing assembly,” such as a spherical element, at the distal ends of their respective gating elements **732**, as described above with reference to FIGS. 2A-2D. In other embodiments, the actuators **730** include sealing assemblies similar to or the same as those described with reference to FIGS. 2A-2D. Regardless of whether the actuators **730** have sealing assemblies or not, the actuators **730** can be configured to selectively control the flow of fluid through the system **700**, as described in further detail below. When the system **700** is assembled, the actuators **730** sit within the corresponding chambers **716** of the first plate **710** and “below” the second plate **715**. The first plate **710**, the second plate **715**, and the actuators **730** are positioned between the first layer **702a** and the second layer **702b**.

[0058] The second layer **702b** can include a first aperture **705a** and a second aperture **705b** (collectively, “the apertures **705**”) for permitting fluid to flow through the second layer **702b** and into the corresponding channels **704**. Each aperture **705** can comprise a corresponding sealing assembly valve **720**, shown as a first sealing assembly valve **720a** and a second sealing assembly valve **720b** (collectively, “the valves **720**”). As best seen in the enlarged cut-away view of the second sealing assembly valve **720b**, the second valve **720b** includes a base portion **722** and a flap portion **724** (which can also be referred to simply as a flap, appendage, moveable feature, etc.). The second aperture **705b** is formed in the base portion **722**. Although described as a base portion **722**, one skilled in the art will appreciate that the base portion **722** can simply be a portion of the second layer **702b** that defines the aperture **705b**. Accordingly, the flap portion **724** can extend from a surface of the second layer **702b**.

[0059] The flap portion **724** can hinge, bend, flex, or otherwise move relative to the base portion **722**. For example, in response to a force applied against the flap portion **724** in the direction A, the flap portion **724** can hinge or otherwise fold in the direction B from a first position (illustrated) in which it does not block the aperture **705** toward a second position (not shown) in which the flap portion **724** is aligned with the base portion **722** and covers or blocks the aperture **705b**. To facilitate the relative motion between the flap portion **724** and the base portion **722**, the valve **720b** can be composed of an at least partially flexible material such as silicone. In other embodiments, the valve **720b** can be made of a generally rigid material with a hinge connecting the flap portion **724** to the base portion **722**. Regardless, the valve **720b** can be biased toward the first position (illustrated) in which it does not block the aperture **705b** (e.g., the valve **720b** generally does not

block the aperture **705b** unless or until an external force moves the flap portion **725** into the second position). In the first position, an angle between the flap portion **724** and the base portion **722** can be between about 60 degrees and about 120 degrees, or between about 70 degrees and about 110 degrees, or between about 80 degrees and about 100 degrees, or about 90 degrees (e.g., the flap portion **724** is generally perpendicular to the base portion **722**). In the second position, an angle between the flap portion **724** and the base portion **722** can be less than about 20 degrees, less than about 10 degrees, less than about 5 degrees, or about 0 degrees (e.g., the flap portion **724** is generally parallel to the base portion **722**).

[0060] In operation, the valves **720** assist with “sealing” or otherwise preventing fluid from flowing through the corresponding apertures **705** when desired. For example, similar to described above with reference to FIG. 2A, the gating elements **732** can be selectively moved by actuating the actuators **730**. To close the valve **720b** and stop fluid flow through the aperture **705b**, the corresponding actuator **730b** can be actuated to move the gating element **732b** in the direction A until the gating element **732b** contacts the flap portion **724** and causes it to bend in the direction B, such that the flap portion **724** blocks the aperture **705b**. To open the valve **720b** and permit fluid flow through the aperture **705b**, the corresponding actuator **730b** can be actuated to move the gating element **732b** in a direction opposite the direction A until the gating element **732b** no longer contacts the flap portion **724**, such that the flap portion **724** resiliently springs back toward the illustrated configuration in which it does not cover the aperture **705b**. In this way, the actuators **730** and the valves **720** operate to selectively control the flow of fluid through the apertures **705**. And because the apertures **705** act as a fluid inlet to the channels **704**, the actuators **730** and the valves **720** can be used to selectively control the flow of fluid through the channels **704**.

[0061] Although the foregoing generally describes the second valve **720b** and the second actuator **730b**, the description applies equally to the first valve **720a** and the first actuator **730a**. Similarly, although the system **700** is shown without a slot, the system **700** can include a slot in addition to the valves **720**. In such embodiments, the slot(s) can further assist with moving the valves into the second (e.g., closed) position by inducing vertical movement of the respective gating element, as described in detail previously. Accordingly, one skilled in the art will appreciate that the embodiments shown and described herein can be combined to form other embodiments with features for improving control of fluid flow through shunts.

[0062] The systems described herein can be designed for shunting fluid between a variety of body regions. For example, in some embodiments the systems described herein are designed to be implanted in a patient's eye to shunt aqueous between the anterior chamber and a target outflow location (e.g., a subconjunctival bleb space), such as to treat glaucoma. Accordingly, in some embodiments the systems described herein can have dimensions compatible with being implanted in the patient's eye. For example, the systems described herein (e.g., the system **100**) may have a length of between about 4 mm and about 20 mm, such as between about 4 mm and 15 mm, or between about 4 mm and 12 mm, or between about 6 mm and 10 mm, or about 8 mm. In some embodiments, the plates described herein (e.g., the plate **110**) can have a width or thickness less than about 500 microns, less than about 400 microns, less than about 300 microns, and/or less than about 200 microns. In some embodiments, the diameter of the fluidic channels and corresponding apertures (e.g., the channels **104** and the inlets **105**) may be less than about 100 microns, less than about 75 microns, and/or less than about 50 microns, such as about 35 microns. Of course, the foregoing dimensions are provided by way of example only, and other dimensions outside the ranges provided above are possible and included within the scope of the present technology. Indeed, the dimensions of the systems described herein may be designed depending on the type of shunting system (e.g., glaucoma shunt vs. hydrocephalus shunt) and intended recipient (e.g., child vs. adult).

EXAMPLES

[0063] Several aspects of the present technology are set forth in the following examples: [0064] 1.

An adjustable shunting system for shunting fluid from a first body region to a second body region within a patient, the system comprising: [0065] a shunting element configured to extend at least partially between the first body region and the second body region of the patient; [0066] a channel extending at least partially through a length of the shunting element between the first body region and the second body region, the channel including an aperture; [0067] a flow control plate having a slot therein, the slot extending between a first end portion having a first width and a second end portion having a second width greater than the first width such that a width of the slot increases from the first end portion to the second end portion, wherein the first end portion is at least partially vertically aligned with the aperture; and [0068] an actuator at least partially extending between the slot and the aperture, wherein the actuator includes a first portion configured to slidably engage the slot and a second portion configured to releasably engage the aperture. [0069] 2. The system of example 1 wherein the actuator is moveable between (a) a first position in which the first portion of the actuator is adjacent the first end portion of the slot and (b) a second position in which the second portion of the actuator is adjacent the second end portion of the slot, and wherein: [0070] in the first position, the second portion of the actuator at least partially blocks fluid flow through the aperture, and [0071] in the second position, the second portion of the actuator does not block fluid flow through the aperture. [0072] 3. The system of example 2 wherein the slot is configured to direct the second portion of the actuator toward the aperture as the actuator is moved from the second position toward the first position. [0073] 4. The system of example 2 or example 3 wherein the slot is configured to direct the second portion of the actuator away from the aperture as the actuator is moved from the first position toward the second position. [0074] 5. The system of any of examples 2-4 wherein the first portion and the second portion of the actuator move (i) in a horizontal direction parallel to a plane of the flow control plate and (ii) a vertical direction normal to the plane of the flow control plate. [0075] 6. The system of any of examples 2-4 wherein the actuator is selectively actuatable to move between the first position and the second position, and wherein the actuator is configured such that, following actuation, the actuator retains its position. [0076] 7 The system of any of examples 1-6 wherein the slot is an indentation in the flow control plate. [0077] 8. The system of any of examples 1-6 wherein the slot is an opening extending between opposing surfaces of the flow control plate. [0078] 9. The system of any of examples 1-6 wherein the slot is formed between one or more ridges extending from a surface of the flow control plate. [0079] 10. The system of any of examples 1-9 wherein the slot is tear-drop shaped. [0080] 11. The system of any of examples 1-10 wherein the slot includes a first side wall and a second side wall, and wherein the first side wall and the second side wall are straight. [0081] 12. The system of any of examples 1-10 wherein the slot includes a first side wall and a second side wall, and wherein the first side wall and the second side wall are curved. [0082] 13. The system of any of examples 1-12 wherein the slot includes one or more frictional elements configured to at least partially restrict movement of the first portion of the actuator within the slot. [0083] 14. The system of any of examples 1-13 wherein the actuator includes a sealing assembly, and wherein the sealing assembly includes the first portion and the second portion. [0084] 15. The system of example 14 wherein the first portion of the sealing assembly and the second portion of the sealing assembly form a single, integral component. [0085] 16. The system of example 14 wherein the sealing assembly has a spherical shape, and wherein the first portion is a first hemispherical portion of the spherical shape and the second portion is a second hemispherical portion of the spherical shape. [0086] 17. The system of example 16 wherein the actuator further comprises a gating element having a cavity extending therethrough, and wherein the sealing assembly is positioned within the cavity and configured to rotate relative to the cavity. [0087] 18. The system of example 14 wherein the first portion of the sealing assembly and the second portion of the sealing assembly are distinct, separate components. [0088] 19. The system of example 18 wherein the first portion is composed of a first material, and wherein the second portion is composed of a second material different than the first material. [0089] 20. The system of example 19 wherein the first material has a higher durometer

than the second material. [0090] 21. The system of any of examples 1-13 wherein the actuator includes a sealing assembly and a gating element, and wherein the sealing assembly includes the first portion and the gating element includes the second portion. [0091] 22. The system of example 21 wherein the actuator includes a recess and the sealing assembly comprises a spherical shape, and wherein the sealing assembly sits at least partially within the recess and is configured to rotate relative to the recess. [0092] 23. The system of any of examples 1-22 wherein the actuator includes a sealing assembly and a gating element, and wherein the sealing assembly is configured to rotate relative to the gating element. [0093] 24. The system of any of examples 1-23 wherein the system is configured to shunt aqueous from an anterior chamber of the patient's eye to a target outflow location within the patient. [0094] 25. An adjustable shunting system for shunting fluid from a first body region to a second body region within a patient, the system comprising: [0095] a shunting element configured to extend at least partially between the first body region and the second body region of the patient, the shunting element having a channel extending at least partially therethrough; [0096] an aperture configured to permit fluid to flow into or out of the channel; [0097] a moveable flap proximate the aperture; and [0098] an actuator configured to selectively move the moveable flap from a first position in which it does not block the aperture to a second position in which it at least partially blocks the aperture. [0099] 26. The system of example 25 wherein the moveable flap extends from a base portion, and wherein the base portion defines the aperture. [0100] 27. The system of example 26 wherein, when the moveable flap is in the first position, an angle between the moveable flap and the base portion is between about 60 degrees and 120 degrees. [0101] 28. The system of example 25 or example 26 wherein, when the moveable flap is in the second position, an angle between the moveable flap and the base portion is less than about 10 degrees. [0102] 29. The system of any of examples 26-28 wherein the moveable flap is perpendicular to the base portion when in the first position and parallel to the base portion when in the second position. [0103] 30. The system of any of examples 25-29 wherein the moveable flap is composed of a flexible material. [0104] 31. The system of example 30 wherein the moveable flap is composed of silicone. [0105] 32. The system of any of examples 26-29 wherein the moveable flap is hingedly coupled to the base portion. [0106] 33. The system of any of examples 25-32 wherein the moveable flap is biased toward the first position. [0107] 34. The system of any of examples 25-33 wherein the actuator and the moveable flap are discrete components. [0108] 35. A method of controlling fluid flow through an implanted shunt, the method comprising: [0109] heating a shape memory actuator above a transition temperature, [0110] wherein heating the shape memory actuator above the transition temperature causes a first portion of the actuator to slide within a slot oriented within a plane parallel to a plane of the actuator, and [0111] wherein the slot causes the first portion to move in both— [0112] a horizontal direction parallel to the plane of the actuator, and [0113] a vertical direction perpendicular to the plane of the actuator. [0114] 36. The method of example 35 wherein heating the shape memory actuator includes (i) heating a first actuation element of the shape memory actuator to cause the first portion to move in both the horizontal direction and the vertical direction toward an aperture, (ii) heating a second actuation element of the shape memory actuator to cause the first portion to move in both the horizontal direction and the vertical direction away from the aperture, or (iii) both (i) and (ii). [0115] 37. The method of example 35 or example 36 wherein the slot causes the first portion of the actuator to simultaneously move in both the horizontal direction and the vertical direction. [0116] 38. The method of example 37 wherein the slot extends between a first end portion having a first width and a second end portion having a second width greater than the first width such that a width of the slot increases from the first end portion to the second end portion.

Conclusion

[0117] The above detailed description of embodiments of the technology are not intended to be exhaustive or to limit the technology to the precise form disclosed above. Although specific embodiments of, and examples for, the technology are described above for illustrative purposes,

various equivalent modifications are possible within the scope of the technology as those skilled in the relevant art will recognize. For example, any of the features of the intraocular shunts described herein may be combined with any of the features of the other intraocular shunts described herein and vice versa. Moreover, although steps are presented in a given order, alternative embodiments may perform steps in a different order. The various embodiments described herein may also be combined to provide further embodiments.

[0118] From the foregoing, it will be appreciated that specific embodiments of the technology have been described herein for purposes of illustration, but well-known structures and functions associated with intraocular shunts have not been shown or described in detail to avoid unnecessarily obscuring the description of the embodiments of the technology. Where the context permits, singular or plural terms may also include the plural or singular term, respectively.

[0119] Unless the context clearly requires otherwise, throughout the description and the examples, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” As used herein, the terms “connected,” “coupled,” or any variant thereof, means any connection or coupling, either direct or indirect, between two or more elements; the coupling of connection between the elements can be physical, logical, or a combination thereof. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application.

Where the context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number respectively. As used herein, the phrase “and/or” as in “A and/or B” refers to A alone, B alone, and A and B. Additionally, the term “comprising” is used throughout to mean including at least the recited feature(s) such that any greater number of the same feature and/or additional types of other features are not precluded. It will also be appreciated that specific embodiments have been described herein for purposes of illustration, but that various modifications may be made without deviating from the technology. Further, while advantages associated with some embodiments of the technology have been described in the context of those embodiments, other embodiments may also exhibit such advantages, and not all embodiments need necessarily exhibit such advantages to fall within the scope of the technology. Accordingly, the disclosure and associated technology can encompass other embodiments not expressly shown or described herein.

Claims

1. An adjustable shunting system for shunting fluid from a first body region to a second body region within a patient, the system comprising: a shunting element configured to extend at least partially between the first body region and the second body region of the patient; a channel extending at least partially through a length of the shunting element between the first body region and the second body region, the channel including an aperture; a flow control plate having a slot therein, the slot extending between a first end portion having a first width and a second end portion having a second width greater than the first width such that a width of the slot increases from the first end portion to the second end portion, wherein the first end portion is at least partially vertically aligned with the aperture; and an actuator at least partially extending between the slot and the aperture, wherein the actuator includes a first portion configured to slidably engage the slot and a second portion configured to releasably engage the aperture.
2. The system of claim 1 wherein the actuator is moveable between (a) a first position in which the first portion of the actuator is adjacent the first end portion of the slot and (b) a second position in which the second portion of the actuator is adjacent the second end portion of the slot, and wherein: in the first position, the second portion of the actuator at least partially blocks fluid flow through the aperture, and in the second position, the second portion of the actuator does not block fluid flow

through the aperture.

3. The system of claim 2 wherein the slot is configured to direct the second portion of the actuator toward the aperture as the actuator is moved from the second position toward the first position.
4. The system of claim 2 wherein the slot is configured to direct the second portion of the actuator away from the aperture as the actuator is moved from the first position toward the first position.
5. The system of claim 2 wherein the first portion and the second portion of the actuator move (i) in a horizontal direction parallel to a plane of the flow control plate and (ii) a vertical direction normal to the plane of the flow control plate.
6. The system of claim 2 wherein the actuator is selectively actuatable to move between the first position and the second position, and wherein the actuator is configured such that, following actuation, the actuator retains its position.
7. The system of claim 1 wherein the slot is an indentation in the flow control plate.
8. The system of claim 1 wherein the slot is an opening extending between opposing surfaces of the flow control plate.
9. The system of claim 1 wherein the slot is formed between one or more ridges extending from a surface of the flow control plate.
10. The system of claim 1 wherein the slot is tear-drop shaped.
11. The system of claim 1 wherein the slot includes a first side wall and a second side wall, and wherein the first side wall and the second side wall are straight.
12. The system of claim 1 wherein the slot includes a first side wall and a second side wall, and wherein the first side wall and the second side wall are curved.
13. The system of claim 1 wherein the slot includes one or more frictional elements configured to at least partially restrict movement of the first portion of the actuator within the slot.
14. The system of claim 1 wherein the actuator includes a sealing assembly, and wherein the sealing assembly includes the first portion and the second portion.
15. The system of claim 14 wherein the first portion of the sealing assembly and the second portion of the sealing assembly form a single, integral component.
16. The system of claim 14 wherein the sealing assembly has a spherical shape, and wherein the first portion is a first hemispherical portion of the spherical shape and the second portion is a second hemispherical portion of the spherical shape.
17. The system of claim 16 wherein the actuator further comprises a gating element having a cavity extending therethrough, and wherein the sealing assembly is positioned within the cavity and configured to rotate relative to the cavity.
18. The system of claim 14 wherein the first portion of the sealing assembly and the second portion of the sealing assembly are distinct, separate components.
19. The system of claim 18 wherein the first portion is composed of a first material, and wherein the second portion is composed of a second material different than the first material.
20. The system of claim 19 wherein the first material has a higher durometer than the second material.
21. The system of claim 1 wherein the actuator includes a sealing assembly and a gating element, and wherein the sealing assembly includes the first portion and the gating element includes the second portion.
22. The system of claim 21 wherein the actuator includes a recess and the sealing assembly comprises a spherical shape, and wherein the sealing assembly sits at least partially within the recess and is configured to rotate relative to the recess.
23. The system of claim 1 wherein the actuator includes a sealing assembly and a gating element, and wherein the sealing assembly is configured to rotate relative to the gating element.
24. The system of claim 1 wherein the system is configured to shunt aqueous from an anterior chamber of the patient's eye to a target outflow location within the patient.
25. An adjustable shunting system for shunting fluid from a first body region to a second body

region within a patient, the system comprising: a shunting element configured to extend at least partially between the first body region and the second body region of the patient, the shunting element having a channel extending at least partially therethrough; an aperture configured to permit fluid to flow into or out of the channel; a moveable flap proximate the aperture; and an actuator configured to selectively move the moveable flap from a first position in which it does not block the aperture to a second position in which it at least partially blocks the aperture.

26. The system of claim 25 wherein the moveable flap extends from a base portion, and wherein the base portion defines the aperture.

27. The system of claim 26 wherein, when the moveable flap is in the first position, an angle between the moveable flap and the base portion is between about 60 degrees and 120 degrees.

28. The system of claim 26 wherein, when the moveable flap is in the second position, an angle between the moveable flap and the base portion is less than about 10 degrees.

29. The system of claim 26 wherein the moveable flap is perpendicular to the base portion when in the first position and parallel to the base portion when in the second position.

30. The system of claim 25 wherein the moveable flap is composed of a flexible material.

31. The system of claim 30 wherein the moveable flap is composed of silicone.

32. The system of claim 26 wherein the moveable flap is hingedly coupled to the base portion.

33. The system of claim 25 wherein the moveable flap is biased toward the first position.

34. The system of claim 25 wherein the actuator and the moveable flap are discrete components.

35. A method of controlling fluid flow through an implanted shunt, the method comprising: heating a shape memory actuator above a transition temperature, wherein heating the shape memory actuator above the transition temperature causes a first portion of the actuator to slide within a slot oriented within a plane parallel to a plane of the actuator, and wherein the slot causes the first portion to move in both—a horizontal direction parallel to the plane of the actuator, and a vertical direction perpendicular to the plane of the actuator.

36. The method of claim 35 wherein heating the shape memory actuator includes (i) heating a first actuation element of the shape memory actuator to cause the first portion to move in both the horizontal direction and the vertical direction toward an aperture, (ii) heating a second actuation element of the shape memory actuator to cause the first portion to move in both the horizontal direction and the vertical direction away from the aperture, or (iii) both (i) and (ii).

37. The method of claim 35 wherein the slot causes the first portion of the actuator to simultaneously move in both the horizontal direction and the vertical direction.

38. The method of claim 37 wherein the slot extends between a first end portion having a first width and a second end portion having a second width greater than the first width such that a width of the slot increases from the first end portion to the second end portion.
